ECON 1550 — Problem Set 5

Spring 2026

Due: March 11, 2026 at 11:59pm ET

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Submission:	Canvas or Gradescope

Instructions

- When submitting to Gradescope, indicate the page where each question is answered to avoid a 5-point deduction.
- Full credit is given for correct answers. If multiple steps are needed, you must show them to get full credit.
- Points are shown for each part. Partial credit is given for partially correct answers; show your work to maximize it.
- Late submissions receive a score of zero.
- If you have technical problems submitting, email your work to the Head TA before the deadline.
- Collaboration with classmates is encouraged; use of generative AI is permitted but discouraged.
- You must write, understand, and submit your solutions individually. Copying other students' or AI-generated answers, even fragments, is not allowed.

1. **The Big Mac Index, Purchasing Power Parity, and the Exchange Rate (100 points)**

The Big Mac index was invented by The Economist in 1986 as a lighthearted guide to whether currencies are at their "correct" level. Read the article about the Big Mac index (attached at the end of this problem set or online at `https://www.economist.com/big-mac-index`), and the box "Some Meaty Evidence on the Law of One Price" in Chapter 5 of the textbook, then please answer the following questions:

(a) [10 points] How is Purchasing Power Parity (PPP) defined in the article from The Economist?

(b) [10 points] How is Purchasing Power Parity (PPP) defined in Chapter 5 of the textbook (and in class)?

(c) [10 points] Measures of exchange rates and PPP for many countries can be downloaded from the OECD at this link. Observations are annual.

Data for the Big Mac index can be found at `https://github.com/TheEconomist/`

big-mac-data. You will use the data in the file big-mac-data/source-data/big-mac-source-data-v2.csv. For some years, observations are annual. For other years, observations are semi-annual. When comparing Big Mac index data to OECD data, transform semi-annual data to annual by taking the average of the two semi-annual observations.

Explain how the OECD PPP measure relates to our definition of PPP in from the textbook (and from class).

Hint: The OECD database links to this explanation of how their measure of PPP is constructed.

(d) [10 points] Pick any two countries that have data for 2023 in both the Big Mac index data and the OECD data. Using these two countries:

For 2023, construct P/P^* using the Big Mac data and the OECD data. Which of the two is closest to E? Do the two P/P^* measures agree on which of the two currencies is over/under-valued?

(e) [10 points] Using all the years available, make a scatter plot of E against the Big Mac data's P/P^*. What should the scatter plot look like if PPP holds? What features of the scatter plot support the hypothesis that PPP holds? What features suggest the hypothesis that PPP holds is not true?

(f) [10 points] Answer the last question once again, but using OECD data to construct P/P^*.

(g) [10 points] Repeat parts e) and f) but now assess relative PPP rather than absolute PPP.

(h) [10 points] In a single plot, show the time series of the growth rate of the two PPP measures and the growth rate of the exchange rate (that is, plot the year in the horizontal axis and the growth rates of the three variables on the vertical axis). Do the series move together over time? Describe one feature of the plot that provides evidence in favor and one feature that provides evidence against the hypothesis that relative PPP holds.

(i) [10 points] Is the PPP implied by the Big Mac data close to the PPP from the OECD data? Give two reasons for why they are not supposed to perfectly agree.

(j) [10 points] There are many criticisms of the Big Mac index as a measure of PPP. The Economist itself points out it is not meant to be a precise measure of PPP or currency valuation. On the other hand, there must be some advantages of the Big Mac index. List three advantages of the Big Mac index.